Solving Linear Equations

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Q1

(i) The first pivot is 2. To eliminate x from the second equation, subtract five times the first equation from the second:

$$\begin{aligned} 2x + 3y &= 1, \\ (10x + 9y) - 5(2x + 3y) &= 11 - 5. \end{aligned}$$

Therefore the upper triangular system is

$$2x + 3y = 1, \qquad -6y = 6.$$

The two pivots are 2 and -6.

(ii) Back substitution gives $y = -1$. The first equation then gives

$$2x + 3(-1) = 1,$$

so $x = 2$. In column form,

$$2\begin{bmatrix} 2 \\ 10 \end{bmatrix} - \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix},$$

which verifies the solution.

(iii) The coefficient matrix and the elimination multiplier are unchanged. For the new right side, the second transformed entry is

$$44 - 5(4) = 24.$$

Thus $-6y = 24$, so $y = -4$. Substitution into $2x + 3y = 4$ gives

$$2x - 12 = 4,$$

and hence $x = 8$. The new solution is $(x, y) = (8, -4)$.

Q2

(i) Since x_1 and x_2 are solutions,

$$Ax_1 = b, \qquad Ax_2 = b.$$

For any $t \in \mathbb{R}$, linearity gives

$$\begin{aligned} A\big((1-t)x_1 + tx_2\big) &= (1-t)Ax_1 + tAx_2 \\ &= (1-t)b + tb = b. \end{aligned}$$

Therefore $(1-t)x_1 + tx_2$ is also a solution.

(ii) Because $x_1 \neq x_2$, the vectors $(1-t)x_1 + tx_2$ are distinct for distinct values of t. Indeed,

$$(1-t_1)x_1 + t_1x_2 = (1-t_2)x_1 + t_2x_2$$

would imply

$$(t_1 - t_2)(x_2 - x_1) = 0,$$

and therefore $t_1 = t_2$. Hence two distinct solutions generate an entire line of solutions. A linear system can have no solution, one solution, or infinitely many solutions, but it cannot have exactly two.

Q3

(i) The multiplier for the entry $a_{21} = 4$ is 4, so

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The multiplier for $a_{31} = -2$ is -2. Subtracting -2 times row 1 is the same as adding twice row 1, so

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}.$$

After these two steps,

$$E_{31}E_{21}A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 4 & 0 \end{bmatrix}.$$

The final multiplier is $4/2 = 2$, and therefore

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

(ii) Multiplying the elimination matrices in the order in which they act gives

$$M = E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix}.$$

Then

$$MA = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix} = U.$$

(iii) The transformed right side is

$$Mb = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ 10 \end{bmatrix}.$$

Thus the triangular system is

$$x + y = 1, \qquad 2y + z = -4, \qquad -2z = 10.$$

Back substitution gives

$$z = -5, \qquad y = \frac{1}{2}, \qquad x = \frac{1}{2}.$$

Therefore

$$x = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -5 \end{bmatrix}.$$

Q4

(i) Direct multiplication gives

$$AB = \begin{bmatrix} 7 & 0 \\ 0 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

Since $AB \neq BA$, these matrices do not commute.

(ii) Distributivity must be used without changing the order of the factors:

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2.$$

(iii) First,

$$A + B = \begin{bmatrix} 2 & 2 \\ 3 & 0 \end{bmatrix}, \qquad (A+B)^2 = \begin{bmatrix} 10 & 4 \\ 6 & 6 \end{bmatrix}.$$

Also $A^2 = A$, $B^2 = B$, and therefore

$$A^2 + 2AB + B^2 = \begin{bmatrix} 16 & 2 \\ 3 & 0 \end{bmatrix}.$$

The two results are different. In general,

$$A^2 + AB + BA + B^2 = A^2 + 2AB + B^2$$

if and only if $BA = AB$. Thus the scalar-style identity is valid precisely when A and B commute.

Q5

(i) Write the rows of A as r_1^T, r_2^T, r_3^T, where

$$r_3^T = r_1^T + r_2^T.$$

If $Ax = e_3$, where $e_3 = (0,0,1)^T$, then its first two equations require

$$r_1^T x = 0, \qquad r_2^T x = 0,$$

while its third equation requires $r_3^T x = 1$. However,

$$r_3^T x = (r_1^T + r_2^T)x = r_1^T x + r_2^T x = 0,$$

which is a contradiction. Thus $Ax = e_3$ has no solution, so A cannot be invertible.
For a general right side b, the same row relation requires

$$b_3 = b_1 + b_2.$$

Indeed, the elimination step $R_3 \leftarrow R_3 - R_1 - R_2$ produces

$$0 = b_3 - b_1 - b_2.$$

If this condition fails, the system is inconsistent. If it holds, the transformed third equation is $0 = 0$ and is redundant. The condition is necessary; further row relations could impose additional conditions.

(ii) Write the columns of A as a_1, a_2, a_3, with $a_3 = a_1 + a_2$. For

$$x = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

we have

$$Ax = -a_1 - a_2 + a_3 = 0.$$

This is a nonzero solution of $Ax = 0$. If A were invertible, multiplication by A^{-1} would imply $x = A^{-1}0 = 0$, a contradiction. Therefore A is not invertible.

Q6

Start with the augmented matrix

$$[A\ I] = \left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right].$$

Apply $R_2 \leftarrow 2R_2$ followed by $R_2 \leftarrow R_2 - R_1$. Then apply $R_3 \leftarrow 3R_3$ followed by $R_3 \leftarrow R_3 - R_2$. These operations give

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 3 & 2 & -1 & 2 & 0 \\ 0 & 0 & 4 & 1 & -2 & 3 \end{array}\right].$$

Apply $R_3 \leftarrow \frac{1}{4}R_3$, then $R_2 \leftarrow R_2 - 2R_3$, and finally $R_2 \leftarrow \frac{1}{3}R_2$:

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{array}\right].$$

Finally, apply $R_1 \leftarrow R_1 - R_2$ and then divide row 1 by 2:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{array}\right].$$

Therefore

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix}.$$

For verification,

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix} = 4I,$$

so $AA^{-1} = I$.

Q7

(i) Subtract row 1 from rows 2 and 3. The first-column multipliers are

$$\ell_{21} = 1, \qquad \ell_{31} = 1.$$

This gives

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \end{bmatrix}.$$

Next subtract twice row 2 from row 3, so $\ell_{32} = 2$. The result is

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Placing the multipliers in their corresponding positions gives

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

Direct multiplication confirms that

$$LU = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix} = A.$$

(ii) The system $Lc = b$ is

$$\begin{aligned} c_1 &= 5, \\ c_1 + c_2 &= 7, \\ c_1 + 2c_2 + c_3 &= 11. \end{aligned}$$

Forward substitution gives

$$c = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}.$$

The system $Ux = c$ is

$$x_1 + x_2 + x_3 = 5, \qquad x_2 + 2x_3 = 2, \qquad x_3 = 2.$$

Back substitution gives

$$x_3 = 2, \qquad x_2 = -2, \qquad x_1 = 5.$$

Thus

$$x = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}.$$

Q8

(i) For $1 \leq i \leq p$ and $1 \leq j \leq m$,

$$\left((AB)^T\right)_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}.$$

On the other hand,

$$\begin{aligned}(B^T A^T)_{ij} &= \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj} \\ &= \sum_{k=1}^{n} b_{ki} a_{jk}.\end{aligned}$$

The two sums are equal because real-number multiplication is commutative. Hence every entry agrees, and

$$(AB)^T = B^T A^T.$$

(ii) The (i, j) entry of $Q^T Q$ is

$$(Q^T Q)_{ij} = q_i^T q_j.$$

Since $Q^T Q = I$,

$$q_i^T q_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Therefore each column has length 1, and distinct columns are orthogonal. Conversely, if the columns are unit vectors and are mutually orthogonal, the same formula shows that every entry of $Q^T Q$ agrees with the corresponding entry of I. Thus $Q^T Q = I$.

(iii) For all $x, y \in \mathbb{R}^n$,

$$(Qx)^T (Qy) = x^T Q^T Q y = x^T y.$$

Thus Q preserves dot products. Taking $y = x$ gives

$$\|Qx\|^2 = (Qx)^T (Qx) = x^T x = \|x\|^2,$$

so Q also preserves lengths. A suitable 2×2 example is the rotation matrix

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Its columns have unit length and zero dot product, so $Q^T Q = I$ and $q_{11} = \cos\theta$.